UniFind

Campus Lost & Found Website

Business Analysis Requirements Document

Field	Details
Project	UniFind – Campus Lost & Found Website
Document Type	Business Analysis Requirements Document
Role	Business Analyst
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1. Project Overview

The UniFind – Campus Lost & Found Website is a web-based platform designed to help university students report lost and found items, search for relevant reports, upload item images, identify possible matches using image matching technology, and communicate with other users regarding potentially recovered items. The website also provides administrative functions for managing users, reports, flags, and disputes.

2. Problem Statement

Students often face difficulties when trying to recover lost belongings within the university environment. Traditional methods such as notices, social media posts, and word-of-mouth can make it difficult to identify matching items and communicate with the correct owner. The proposed website provides a centralized digital platform where students can report, search, match, and communicate about lost and found items.

3. Project Objectives

- Allow students to report lost items.
- Allow students to report found items.
- Provide an easy way to search for lost and found items.
- Allow users to upload item images.
- Identify possible matching items using image matching.
- Allow users to communicate about possible matches.
- Allow users to flag inappropriate content or behavior.
- Allow administrators to manage users, reports, flags, and disputes.
- Provide a reliable, responsive, and user-friendly campus lost-and-found website.

4. Stakeholders

Stakeholder	Role / Interest
Students	Main users who report, search, match, and recover lost or found items.
System Administrator	Manages users, reports, flags, disputes, and administrative actions.
Project Team	Analyzes, designs, develops, tests, and maintains the website.
University	Provides the campus environment, policies, and governance context.
Business Analyst	Elicits, analyzes, documents, prioritizes, and validates requirements.
QA / Tester	Verifies that implemented website functionality satisfies acceptance criteria.

5. System Actors

Actor	Responsibilities
Student / User	Register, log in, manage profile, report items, upload images, search, view matches, communicate, flag content, submit disputes, and provide feedback.

Administrator	Manage users and reports, review flags and disputes, and take appropriate administrative action.
Image Matching System	Analyze and compare item images and return possible matches with similarity information.
Notification System	Deliver relevant notifications such as possible matches, messages, and status updates.
Web Browser	Provides the interface through which users access the web-based website.

6. Project Scope

6.1 In Scope

- User registration and authentication
- User profile management
- Lost item reporting
- Found item reporting
- Item image upload and validation
- Item search and filtering
- Item details and status management
- Image-based matching
- Possible match results
- Communication between authorized users
- Notifications
- Flagging users or reports
- Dispute submission and resolution
- Reviews / feedback
- Administrative management

6.2 Out of Scope

- Physical delivery of recovered items
- Buying or selling items
- Payment processing
- External marketplace integration
- Advanced facial recognition
- Automatic physical verification of ownership

6.3 Platform and Access

The system will be developed as a web-based website. Users will access the website through a modern web browser using an internet connection. The website will provide responsive and user-friendly interfaces suitable for desktop and mobile screen sizes.

7. Functional Requirements

ID	Functional Requirement	Priority
FR-01	The system shall allow students to create a UniFind account by providing their name, university email address, password, and password confirmation. The system shall validate the entered information and prevent duplicate registrations.	Must
FR-02	The system shall allow registered users to log in and log out.	Must
FR-03	The system shall allow users to manage their profiles.	Should
FR-04	The system shall allow users to report a lost item.	Must
FR-05	The system shall allow users to report a found item.	Must
FR-06	The system shall allow users to upload images of lost/found items.	Must
FR-07	The system shall allow users to edit and delete their own item reports.	Must
FR-08	The system shall display lost and found item listings.	Must
FR-09	The system shall allow users to search for items by keywords.	Must
FR-10	The system shall allow users to filter items by category, location and date.	Should
FR-11	The system shall compare uploaded item images with existing item images.	Must
FR-12	The system shall display possible image matches ranked by similarity.	Must
FR-13	The system shall allow users to view detailed information about an item.	Must
FR-14	The system shall allow users to contact another user regarding a possible match.	Should
FR-15	The system shall allow users to send and receive messages.	Should
FR-16	The system shall notify users about relevant updates or possible matches.	Should
FR-17	The system shall allow users to flag inappropriate reports or users.	Should
FR-18	The system shall allow users to submit disputes.	Should
FR-19	The system shall allow users to provide reviews/feedback after an interaction.	Could
FR-20	The system shall allow administrators to manage users.	Must
FR-21	The system shall allow administrators to review flagged content/accounts.	Must
FR-22	The system shall allow administrators to review and resolve disputes.	Must
FR-23	The system shall allow administrators to warn, suspend or deactivate accounts.	Must
FR-24	The system shall maintain item statuses such as Lost, Found, Matched, Returned and Closed.	Must

8. Non-Functional Requirements

ID	Non-Functional Requirement	Priority
NFR-01	Normal user requests should receive a response within approximately 2–3 seconds under normal conditions.	Should
NFR-02	Image matching should provide results within an acceptable time.	Should
NFR-03	User passwords must be securely hashed and never stored as plain text.	Must
NFR-04	Protected features must require authentication.	Must
NFR-05	Users must only be able to modify their own reports.	Must
NFR-06	Private user information and conversations must only be accessible to authorized users.	Must

NFR-07	The system should minimize unexpected failures and data loss.	Must
NFR-08	The website interface should be simple, responsive, consistent, and easy for students to navigate on desktop and mobile devices.	Must
NFR-09	The website should work correctly on supported modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.	Should
NFR-10	The system should maintain accurate and consistent data.	Must
NFR-11	Important administrative actions should be recorded.	Should
NFR-12	Uploaded images should be validated for supported format and size.	Must
NFR-13	The system should display clear error messages when invalid information is entered.	Must
NFR-14	The system should be maintainable and structured for future development.	Should

9. Business Rules

Rule ID	Business Rule
BR-01	Only students who meet the agreed university email requirements may register for a UniFind account and Only registered users can submit lost or found item reports.
BR-02	A user can edit or delete only their own reports.
BR-03	An item report must contain all required information before submission.
BR-04	An item may have a status of Lost, Found, Possible Match, Matched, Returned, or Closed.
BR-05	A user must not be able to access another user's private conversations.
BR-06	Administrators can review reports that have been flagged.
BR-07	Administrators can take action against accounts that violate system rules.

10. Flagging Rules

A user or item report may be flagged when:

- A user submits false information.
- A user repeatedly creates misleading reports.
- A user uploads inappropriate content.
- A user harasses another user.
- A user behaves abusively.
- A user attempts fraudulent activity.
- A user violates applicable university rules.

Flagging Process:

1. User submits flag
2. System records flag

3. Administrator reviews flag
4. Administrator investigates
5. Administrator makes a decision
6. Action is taken if required
7. Flag is resolved

11. Administrative Actions

- Review flagged users.
- Review flagged reports.
- Review disputes.
- Send warnings.
- Temporarily suspend accounts.
- Permanently deactivate accounts.
- Remove inappropriate reports.
- Resolve disputes.

12. Dispute Management

A user may submit a dispute when:

- They believe an ownership claim is incorrect.
- An item was not returned as expected.
- Another user provided false information.
- There was inappropriate behavior.
- There is disagreement regarding a reported item.

Dispute lifecycle:

Status	Meaning
Open	Dispute has been submitted and is awaiting review.
Under Review	Administrator is investigating the dispute and evidence.
Resolved	Administrator has completed the review and recorded a decision.

Possible outcomes:

- No Action
- Warning
- Account Suspension
- Account Deactivation
- Report Removed

13. User Story Backlog Summary

ID	Epic	User Story	Priority
US-01	User Management	Register Account	Must
US-02	User Management	Login	Must
US-03	User Management	Logout	Must
US-04	User Management	Manage Profile	Should
US-05	Lost & Found Item Management	Report Lost Item	Must
US-06	Lost & Found Item Management	Report Found Item	Must
US-07	Lost & Found Item Management	Upload Item Image	Must
US-08	Lost & Found Item Management	View Item Details	Must
US-09	Lost & Found Item Management	Edit Item Report	Must
US-10	Lost & Found Item Management	Delete Item Report	Must
US-11	Lost & Found Item Management	Update Item Status	Must
US-12	Image Matching	Upload Image for Matching	Must
US-13	Image Matching	Compare Item Images	Must
US-14	Image Matching	Display Possible Matches	Must
US-15	Image Matching	View Match Details	Must
US-16	Search & Discovery	Search Items	Must
US-17	Search & Discovery	Filter Items	Should
US-18	Search & Discovery	View Search Results	Must
US-19	Communication & Notifications	Contact User	Should
US-20	Communication & Notifications	Send Message	Should
US-21	Communication & Notifications	Receive Message	Should
US-22	Communication & Notifications	Receive Notifications	Should
US-23	Reviews & Feedback	Submit Feedback	Could
US-24	Reviews & Feedback	View Feedback	Could
US-25	Administration & Disputes	Manage Users	Must
US-26	Administration & Disputes	Review Item Reports	Must

US-27	Administration & Disputes	Review Flagged Accounts	Must
US-28	Administration & Disputes	Flag User or Report	Should
US-29	Administration & Disputes	Submit Dispute	Should
US-30	Administration & Disputes	Review Dispute	Must
US-31	Administration & Disputes	Resolve Dispute	Must
US-32	Administration & Disputes	Suspend User	Must
US-33	Administration & Disputes	Deactivate User	Must

14. Sprint 1 Candidate Stories

The following stories are recommended as Sprint 1 candidates. Final Sprint 1 scope should be agreed with the development team and project stakeholders.

- Register Account
- Login
- Report Lost Item
- Report Found Item
- Upload Item Image
- View Item Listings
- Search Items
- View Item Details
- Basic Image Matching

15. Requirement Traceability

Requirement(s)	Epic	Representative Stories	Business Area
FR-01 / FR-02	User Management	Register Account / Login	Account access
FR-04 / FR-05	Lost & Found Item Management	Report Lost Item / Report Found Item	Item reporting
FR-06 / FR-11 / FR-12	Image Matching	Upload / Compare / Display Matches	Image matching
FR-09 / FR-10	Search & Discovery	Search / Filter Items	Discovery
FR-14 / FR-15 / FR-16	Communication & Notifications	Contact / Message / Notifications	Communication
FR-17 / FR-18	Administration & Disputes	Flag / Submit Dispute	Governance
FR-20–FR-24	Administration & Disputes	Manage / Review / Resolve	Administration

16. Assumptions and Constraints

16.1 Assumptions

- The primary users are university students.
- Users have access to a modern web browser and internet connection.
- Users provide reasonably accurate item information.
- The image matching functionality can be implemented using an appropriate technical approach.
- Administrative users are authorized by the university/project context.

16.2 Constraints

- Limited project time and resources.
- Image quality and the selected image-matching approach may affect matching accuracy.
- University privacy, security, and usage policies must be considered.
- The project focuses on core campus lost-and-found functionality rather than a general marketplace.
- The website must remain usable across supported modern browsers and common screen sizes.

Requirements should be reviewed and agreed before development. Changes after approval should be documented, assessed for impact, and communicated to the project team.

17. Requirements Approval

The requirements should be reviewed and agreed by the project team before implementation.

Role	Responsibility	Name	Approval / Status	Date
Project Manager	Reviews and approves project scope and requirements	P. D. M. Wanigasekara	Approved	07 October 2026
Business Analyst	Prepares, documents, and maintains project requirements	M. A. P. S. Perera	Approved	06 October 2026
QA Engineer	Reviews requirements for clarity, completeness, and testability	K. M. D. Tishani	Approved	07 October 2026
Developer 1	Reviews requirements for technical feasibility	W. D. Sampath	Approved	07 October 2026
Developer 2	Reviews requirements for technical feasibility	H. D. M. Rukshani	Approved	07 October 2026

Developer 3	Reviews requirements for technical feasibility	W. R. A. P. Sakura	Approved	07 October 2026